

任务生命周期泳道图：FSM、HITL 与恢复闭环

```

graph TD
    subgraph Human_UI [Human / UI]
        Submit([Submit...])
        Review([Review candidates...])
        Approve([Approve / Replan...])
        View([View report...])
        Failed([FAILED / CANCELLED...])
    end

    subgraph TaskAPI [TaskAPI]
        Validate[Validate schema...]
    end

    subgraph Workflow_FSM [Workflow / FSM]
        Created[CREATED -> PLANNING]
        Planned[PLANNED...]
        Running[RUNNING...]
        Step{Step...}
        Waiting[WAITING_PATCH_CONFIRM...]
        Summarizing[SUMMARIZING]
    end

    subgraph PlannerAgent [PlannerAgent]
        GenerateTopK[Generate Top-K...]
        GeneratePatch[Generate Patch / ...]
    end

    subgraph Executor_Tools [Executor / Tools]
        Retire[Retire / local failure...]
        Summarizer[SummarizerAgent...]
    end

    subgraph SafetyAgent [SafetyAgent]
        Safety{Safety...}
    end

    subgraph Storage_Log [Storage / Log]
        PersistPending[Persist PendingAction...]
        WriteEventLog[Write EventLog...]
        PersistReport[Persist report...]
    end

    Submit --> Validate
    Validate --> Created
    Created -- "planning request" --> GenerateTopK
    GenerateTopK -- "CandidateSet" --> Plan{Plan...}
    Plan -- "Yes: wait" --> PersistPending
    Plan -- "No: auto" --> Planned
    Planned --> Running
    Running -- "pre/post check" --> Safety
    Safety -- "allow / warn" --> Retire
    Safety -- "block / replan" --> GeneratePatch
    Retire -- "patch request" --> GeneratePatch
    GeneratePatch -- "Patch/Replan candidates" --> Waiting
    Waiting -- "all steps done" --> Summarizing
    Waiting -- "failure / retry exhausted" --> Step
    Step -- "more steps" --> GeneratePatch
    Step -- "resume RUNNING" --> Running
    Summarizing --> Summarizer
    Summarizer -- "reports / artifacts" --> PersistReport
    PersistReport -- "timeline + report" --> View
    PersistReport -- "cancel / terminal_stop" --> Failed
    Approve -- "accept" --> Planned
    Approve -- "HITL / WAITING" --> Summarizing
    PersistPending -.- "expose PendingAction" --> Review
    Review -.- "HITL / WAITING" --> Approve
    WriteEventLog -.- "HITL / WAITING" --> Approve
  
```

Legend:

- HITL / WAITING
- Planning
- Execution
- Persistence
- HITL / WAITING

Invariant: WAITING_* states are paused; TaskSnapshot/EventLog are written before human Decision is requested.

